

Query Intent Resolver Project Outline

Project Goal

Build an intelligent routing system that understands every user query, classifies the users intent (and persona), and short circuits simple requests while sending complex ones to more advanced workflows, minimizing latency and cost while maximizing accuracy and user experience. The system acts as the textual front end router for search across our app and website.

Success Criteria

By the end of this phase we should have:

1. Defined intent taxonomy
2. Defined persona taxonomy
3. Thousands of labeled queries in CSV format
4. Clear easy vs complex routing strategy
5. Initial classifier design
6. Dataset ready for Python ingestions
7. Simple evaluation plan with sample tests

If it cannot be exported and used immediately by engineering → it's not done.

Phase 1- Persona & Behavior Research

Objective: Understand who is querying before deciding how to classify.

Deliverable: Persona Doc

Define core personas:

- High school students
- College students

- Parents
- counselors/teachers
- Advisors
- Colleges (B2B)

Other personas:

- Transfer student
- Community college student
- International student
- Graduate applicant
- Career changer
- Employer / Recruiter
- Industry hiring manager
- EdTech founder
- Scholarship foundation officer
- Government workforce analyst

For each:

- Goals
- Typical questions
- Vocabulary style
- Complexity level
- Example queries

Why: Intent often depends on persona

Example:

- Scholarships → HS students
- Tour Schedule → parent
- Yield Rate → Counselor

Phase 2- Intent Taxonomy Design

Objective: Define the buckets queries will route into.

Deliverable: Intent Definition Doc

For each intent:

- Name
- Description
- Examples
- Required parameters/entities
- Expected pipelines
- Complexity (easy/medium/complex)

Example:

direct_lookup → “MIT”, “UCLA tuition” → Route: Typesense → school page → short circuit

Multi_criteria_filter → “affordable engineering schools in California with good job placement”

→ Route: multi-step agent workflow

Guidance:

- 8-15 categories max
- Must map directly to routing logic
- Avoid over granularity

Phase 3- Query Dataset Creation

Objective: Generate thousands of realistic, labeled queries.

Format (Required):

Google sheet → CSV export

| query_text | persona | intent_label | complexity | entities | route | notes |

Requirements:

- 3,000-5,000+ queries minimum
- 200-400 per intent/persona combo
- Include:
 - Slang
 - Typos
 - Incomplete queries
 - Messy phrasing
 - Conversational follow ups

Generation Sources:

- LLM Prompts
- Persona simulation
- Real world examples
- Manual brainstorming

Think training data.

Phase 4- Clustering & Pattern Discovery

Objective: Use data science + LLMs to validate taxonomy

Tasks:

- Cluster queries (K-means / embeddings / LLM grouping)
- Identify natural buckets
- Compare to manual taxonomy
- merge/split categories if needed

Deliverable:

- Short findings doc
 - What clusters emerged?
 - What surprised us?
 - Taxonomy changes?

Phase 5- Routing & Short Circuit Strategy

Objective: Design fast/cheap vs slow/complex paths

Core Principle: Latency + token cost optimization

Define:

Easy → cheap model or direct search

Medium → lightweight LLM

Complex → multi agent workflow

Example:

Intent	Path
direct_lookup	Typesense only
simple_filter	Cheap classifier
complex_query	Foundry multi-agents

Deliverable: Routing decision tree diagram

Phase 6- Classifier Design

Objective: Specify how classification happens.

Define:

- Input: query text
- Output
 - Persona
 - Intent
 - Confidence
 - Entities

Approach options:

- LLM prompt considerations
- Cost considerations
- Fallback behavior
- Confidence thresholds

Deliverable:

- 1-2 page design doc

Phase 7- Evaluating Plan

Objective: ensure it actually works.

Steps:

- Hand label 2–0300 sample queries
- Test classifier
- Measure:
 - Accuracy

- Latency
- Token cost
- Correct short circuit rate

Foundry will track:

- Cost
- Time
- Routing path

Deliverable:

- Simple metrics dashboard or report

Weekly Milestone Plan

Week 1: Persona definitions + draft intents

Week 2: Dataset scaffold + 1,000 queries

Week 3: 3,000+ queries + clustering

Week 4: Final dataset + routing design + classifier draft

Week 5: Evaluation + Iteration